

HANDS-ON LAB EXERCISE 01

Create GitHub Account, Generate Token, Create Repository, Push, Pull and Clone Code

Training Information

Training Name: GitHub Fundamentals for Beginners

Trainer: Sandeep Kumar Sharma

Duration: 90 Minutes

Difficulty Level: Beginner

Lab Type: Hands-On Practical Exercise

Learning Objective

By the end of this lab, participants will be able to:

- Understand Git and GitHub fundamentals
 - Create a GitHub account
 - Generate a Personal Access Token (PAT)
 - Create and manage GitHub repositories
 - Initialize a local Git repository
 - Connect local repositories with GitHub
 - Push code from local machine to GitHub
 - Clone repositories from GitHub
 - Pull latest changes from GitHub
 - Track source code changes using Git
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Learning Outcome

After successful completion of this lab, participants will be able to:

- Work with GitHub repositories independently
- Upload source code to GitHub
- Download repositories from GitHub
- Synchronize local and remote repositories
- Perform basic version control operations

- Collaborate with development teams using GitHub
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Prerequisites

- Basic computer knowledge
 - Internet connectivity
 - GitHub account (can be created during lab)
 - Git installed on local machine
 - Visual Studio Code (Recommended)
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Lab Requirements

Hardware

- Laptop/Desktop
- Minimum 4 GB RAM

Software

- Git
 - Web Browser
 - Visual Studio Code (Optional)
-

Introduction

What is Git?

Git is a Distributed Version Control System (DVCS) used to track changes in source code and maintain version history.

Using Git, developers can:

- Track code changes
 - Roll back changes
 - Collaborate with team members
 - Manage software versions
-

What is GitHub?

GitHub is a cloud-based platform used to host Git repositories.

GitHub helps developers:

- Store code online
- Share projects
- Collaborate with teams
- Manage versions and releases

Git vs GitHub

Git	GitHub
Version Control Tool	Cloud Hosting Platform
Installed Locally	Available Online
Tracks Changes	Stores Repositories
Command Line Tool	Web-Based Interface

GitHub Workflow

```
graph TD; A[Create Account] --> B[Generate Token]; B --> C[Create Repository]; C --> D[Create Local Project]; D --> E[Git Init]; E --> F[Git Add]; F --> G[Git Commit]; G --> H[Git Remote Add Origin]; H --> I[Git Push]; I --> J[Git Clone]; J --> K[Git Pull];
```

Exercise 1: Create GitHub Account

Step 1

Open browser and navigate to:

<https://github.com>

Step 2

Click:

Sign Up

Provide:

- Email Address
- Username
- Password

Complete verification process.

Step 3

Verify email address and login successfully.

Exercise 2: Generate Personal Access Token (PAT)

Why Do We Need a Token?

GitHub no longer allows password authentication for Git operations.

Instead of password, we use:

- Username
 - Personal Access Token (PAT)
-

Step 1

Click:

Profile → Settings

Step 2

Navigate to:

Developer Settings

Step 3

Select:

Personal Access Tokens

Then:

Tokens (Classic)

Step 4

Click:

Generate New Token

Step 5

Provide:

Token Name : GitHub Lab Token
Expiration : 90 Days

Select Permissions:

- ✓ repo
- ✓ workflow
- ✓ read:org

Step 6

Click:

Generate Token

Step 7

Copy the generated token.

Example:

ghp_XYZ123ABC456DEF789

Save it securely.

Important: The token will be displayed only once.

Exercise 3: Install Git

Windows

Download Git from:

<https://git-scm.com>

Install using default settings.

Verify Installation

Open Command Prompt:

git --version

Expected Output:

```
git version 2.x.x
```

Exercise 4: Configure Git

Configure Username:

```
git config --global user.name "Your Name"
```

Configure Email:

```
git config --global user.email "yourmail@example.com"
```

Verify Configuration:

```
git config --list
```

Exercise 5: Create Local Project

Create project directory:

```
mkdir github-lab  
cd github-lab
```

Create sample file:

```
echo "Welcome to GitHub Lab" > index.html
```

Verify:

```
dir
```

or

```
ls
```

Exercise 6: Initialize Local Git Repository

Initialize repository:

```
git init
```

Expected Output:

```
Initialized empty Git repository
```

Exercise 7: Check Repository Status

```
git status
```

Expected Output:

```
Untracked files:  
index.html
```

Exercise 8: Add Files to Staging Area

```
git add .
```

Verify:

```
git status
```

Expected Output:

```
Changes to be committed
```

Exercise 9: Commit Changes

```
git commit -m "Initial Commit"
```

Expected Output:

```
1 file changed
```

Exercise 10: Create GitHub Repository

Login to GitHub.

Click:

New Repository

Provide:

```
Repository Name : github-lab  
Description : My First GitHub Repository
```

Important

Do NOT select:

```
README  
.gitignore  
License
```

Click:

Create Repository

Exercise 11: Connect Local Repository to GitHub

GitHub will display repository URL.

Example:

```
https://github.com/username/github-lab.git
```

Add remote repository:

```
git remote add origin https://github.com/username/github-lab.git
```

Verify:

```
git remote -v
```

Expected Output:

```
origin https://github.com/username/github-lab.git  
origin https://github.com/username/github-lab.git
```

Exercise 12: Push Code to GitHub

For Main Branch:

```
git push -u origin main
```

For Master Branch:

```
git push -u origin master
```

Authentication

GitHub will ask:

```
Username:
```

Enter:

```
GitHub Username
```

Then:

Password:

Paste:

Personal Access Token (PAT)

NOT your GitHub password.

Expected Output

```
Enumerating objects...  
Writing objects...  
Branch main set up to track origin/main.
```

Refresh GitHub repository.

Verify file upload successfully.

Exercise 13: Modify Existing File

Open:

index.html

Update content:

```
<h1>GitHub Lab Updated</h1>
```

Save file.

Exercise 14: Commit Updated Changes

Check status:

```
git status
```

Add changes:

```
git add .
```

Commit changes:

```
git commit -m "Updated HTML File"
```

Exercise 15: Push Updated Changes

```
git push
```

Refresh GitHub repository.

Verify latest changes.

Exercise 16: Clone Repository

Open a new folder.

Execute:

```
git clone https://github.com/username/github-lab.git
```

Expected Output:

```
Cloning into 'github-lab'...
```

Move inside folder:

```
cd github-lab
```

Verify files downloaded successfully.

Exercise 17: Make Changes Directly from GitHub

Login to GitHub.

Edit:

```
README.md
```

Add:

```
This repository is created during GitHub Lab.
```

Save changes.

Exercise 18: Pull Latest Changes

On local machine execute:

```
git pull origin main
```

Expected Output:

```
Latest changes downloaded successfully.
```

Exercise 19: View Commit History

```
git log
```

Expected Output:

```
Commit ID
Author
Date
Commit Message
```

Important Git Commands

Command	Description
git init	Initialize Repository

Command	Description
git status	Check Repository Status
git add .	Add Files
git commit -m	Create Commit
git remote add origin	Connect GitHub Repository
git remote -v	Verify Remote Repository
git push -u origin main	First Push
git push	Upload Changes
git clone	Clone Repository
git pull	Download Latest Changes
git log	View Commit History

Challenge Activity

Perform the following tasks independently:

1. Create a new GitHub repository.
2. Create a local project.
3. Initialize Git repository.
4. Create a new HTML file.
5. Commit changes.
6. Connect repository using git remote add origin.
7. Push code to GitHub.
8. Clone repository on another folder.
9. Make changes from GitHub portal.
10. Pull latest changes locally.

Lab Validation Checklist

Task	Completed
GitHub Account Created	☐
Personal Access Token Generated	☐
Git Installed	☐
Local Repository Created	☐
Git Initialized	☐

Task	Completed
Commit Created	☐
GitHub Repository Created	☐
Remote Repository Added	☐
First Push Successful	☐
Repository Cloned	☐
Pull Successful	☐
Commit History Verified	☐

Lab Summary

In this lab, participants learned how to:

- Create a GitHub account
- Generate Personal Access Token (PAT)
- Install and configure Git
- Create local repositories
- Initialize Git repositories
- Create commits
- Create GitHub repositories
- Connect local repositories to GitHub
- Push code to GitHub
- Clone repositories
- Pull latest changes
- View commit history

This lab provides a complete beginner-to-intermediate GitHub workflow that every developer uses in real-world software projects.

Trainer: Sandeep Kumar Sharma

Lab Status: Successfully Completed 